

Syed Abubaker Ahmed

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DOB - 25/06/2003 | Citizenship – India | Iqama – Transferable | Driving License - Available



PROFESSIONAL SUMMARY

A dedicated Computer Science engineer focused on backend development with hands-on experience in building scalable applications and REST APIs. I am skilled in Java and Python and backend frameworks and I can build reliable and efficient data driven systems. I learn fast, solve problems with clarity and consistently deliver high quality backend solutions.

Explore my full portfolio and interactive project demos at syed-abubaker-ahmed.github.io

EXPERIENCE

Backend Support Engineer

MINE

Apr 2025 – Present

Remote / USA

- Design and tested 20+ Spring Boot REST APIs across multiple service endpoints with a strong focus on request validation response consistency error handling and smooth integration within the platform's full-stack architecture. This work ensures reliable backend behavior during active development and internal testing phases.
- Debug and resolve 18+ recurring backend and API issues by closely analyzing application logs stack traces and failed API responses. These fixes address integration flaws data handling inconsistencies and stability problems that directly affect system reliability.
- Collaborate closely with frontend developers to support backend and frontend integration and feature delivery. This includes refining API contracts aligning request and response structures and resolving data flow issues to ensure seamless communication between application layers.
- Built and implemented automated API testing scripts to reduce manual testing effort and improve testing consistency across services. These scripts support regression testing validate API behavior after changes and help detect backend issues earlier in the development cycle.
- Maintain clear and structured backend documentation that details API endpoints request parameters response formats and known behaviors. This documentation supports efficient development workflows improves cross team understanding and enables faster onboarding for contributors.
- Developed an internal quiz generator tool using Spring Boot principles to automate content transformation, reducing manual quiz creation work and demonstrating backend system design from data ingestion to structured output.
- Consistently complete assigned backend tasks within expected timelines while working part-time and adapting quickly to changing development requirements.

Key Achievements:

- Strengthened backend stability by improving API behavior and response handling which resulted in a measurable increase in response reliability during internal testing.
- Reduced recurring backend failures through consistent debugging and targeted fixes which improved overall system reliability and lowered repeated error occurrences.
- Developed and integrated a robust backend service using Java and Maven that reliably transforms article content into multi-question quizzes. Implementing structured error handling and JSON data modeling for production use.

EDUCATION

VIT Vellore (Vellore Institute of Technology)

Bachelor of Technology in Computer Science and Engineering

2021 – 2025

Vellore, India

PROJECTS

Smart Search POC | *Java, Spring Boot, AWS Architecture*

Jan 2026 – Feb 2026

- Designed and implemented an AI-powered search proof-of-concept with hybrid ranking logic and automatic editorial alignment scoring eliminating the need for manual content tagging.
- Built algorithmic guardrails to filter misaligned content in real time using multi-factor scoring enforcing alignment thresholds and source prioritization rules.
- Architected a scalable production migration strategy leveraging AWS services defining end-to-end ingestion search scoring pipelines and cost-optimized secure cloud deployment patterns.

Youtube-Shorts-Uploader | *Python, ComfyUI, XTTS v2*

Jan 2026 – Feb 2026

- Designed and implemented a fully automated short-form video generation pipeline orchestrating Groq LLM for ideas and scripts, ComfyUI for image generation, XTTS v2 for voice synthesis and FFmpeg for vertical video assembly.
- Built a production-ready orchestration layer with SQLite-backed state tracking, job recovery, and fault-tolerant retries, enabling continuous hands-off content generation and resumability after failures.
- Engineered end-to-end video creation producing 1080×1920 MP4 outputs with synchronized narration, captions, and visuals, alongside optional randomized auto-upload scheduling for scalable YouTube Shorts publishing.

Crash Detection System | *YOLOv8, CNN, OpenCV*

Apr 2025 – Jun 2025

- Built and managed a backend-driven accident detection pipeline that combined model inference with custom dataset construction, achieving 97.5% accuracy, 98.5% precision and 96.6% recall through reliable data ingestion validation and processing workflows.
- Designed backend validation logic using multi frame crash confirmation and confidence-state handling to improve detection reliability, reducing false positives to 1.5% while sustaining ~0.87 ms/frame inference latency at the service level.
- Implemented cloud based backend integrations to handle confirmed crash events by uploading incident images and triggering automated WhatsApp SMS and call alerts via external APIs, strengthening real-time emergency responsiveness.

Quiz Generator | *Java, Groq API, Maven, Jackson*

Jan 2026 – Jan 2026

- Engineered a production ready backend service that automates quiz generation from articles, integrating Groq's Llama 3.3 70B API for question creation and implementing structured JSON data handling with retry logic and model fallback.
- Built a multi stage processing pipeline for article ingestion, HTML text cleaning, token estimation and structured quiz output. Delivering consistent 20-question quizzes with proper difficulty distribution (8 Easy, 7 Medium, 5 Hard).
- Designed a maintainable Java architecture with clear separation of concerns (Article, Quiz, Question models), configurable API parameters and comprehensive error handling to ensure reliable operation and smooth integration within development workflows.

TECHNICAL SKILLS

Languages: Java, Python, SQL, C/C++

Backend Frameworks: Spring Boot, Node.js, Express.js, Jackson

Data and ML Integration: TensorFlow/Keras, YOLOv8, OpenCV, aioquic

Developer Tools: Postman, Git, Docker, VS Code, IntelliJ IDEA

Platforms: AWS, GCP